

On the Road to Productivity: Investigating Text-Presentation Techniques and Audio Assistance for Non-Driving Tasks in Conditionally Automated Vehicles

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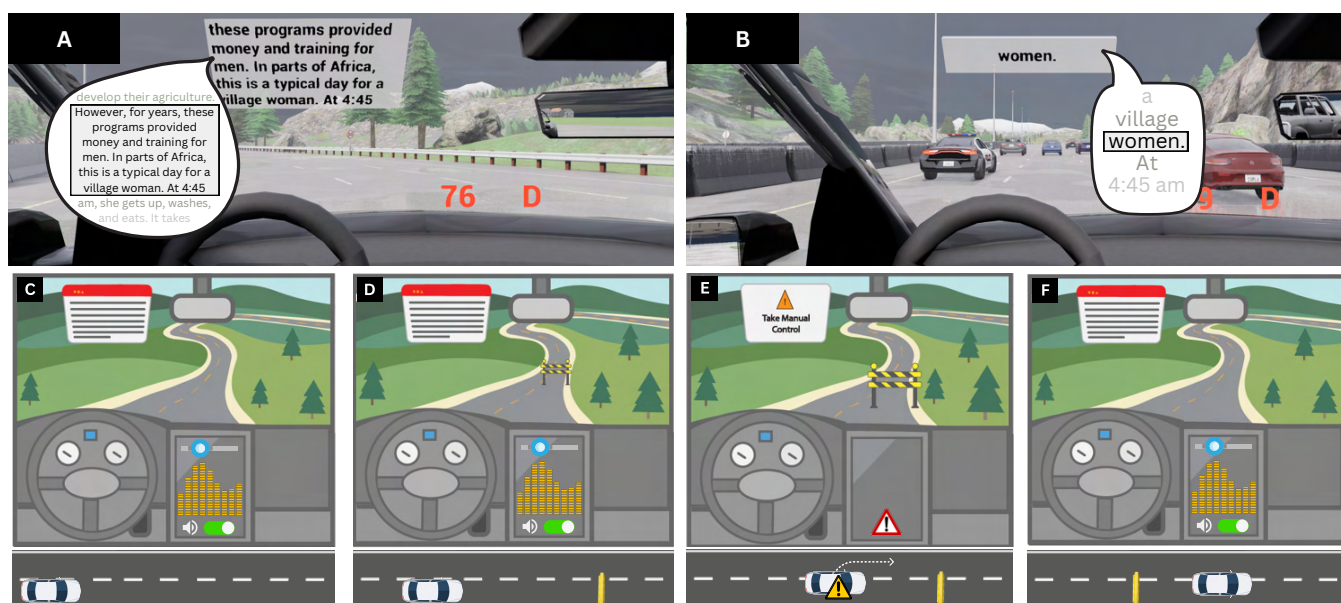


Figure 1: We evaluate reading interfaces to support non-driving related reading tasks in conditionally automated vehicles using two text-presentation techniques ((A) vertical scrolling text presentation [3], and (B) rapid serial visual presentation [14]) in combination with (C, D) auditory speech displays. (C, D) Our envisioned scenario has the driver perform a non-driving-related task (NDRT) of reading while the vehicle is engaged with driving automation. (E) The NDRT is paused when a take-over request is issued by the vehicle, requiring the driver to take manual control. (F) The driver returns to the NDRT once normal conditions are restored and the vehicle has re-engaged driving automation.

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ABSTRACT

Conditionally automated vehicles provide unique opportunities for drivers to engage in non-driving-related tasks (NDRTs); however, drivers must remain prepared to respond to take-over requests. This paper explores design challenges and potential solutions for supporting reading as an NDRT in SAE Level 3 vehicles. Specifically, we assess two prominent text-presentation techniques: vertical scrolling text presentation (VSTP) and rapid serial visual presentation (RSVP), exploring both in conjunction with their integration

with auditory speech displays (ASD). A driving simulation study involving $N = 32$ participants revealed that RSVP surpassed VSTP in regaining situational awareness, as indicated by lower average braking actuation, and was also preferred by participants. The integration of ASDs with both techniques reduced perceived cognitive workload and improved the user experience, albeit with compromised lateral control. Our findings can help advance the design of human-centered interfaces for reading in conditionally automated vehicles.

CCS CONCEPTS

• **Human-centered computing** → **Interaction devices**.

KEYWORDS

Conditionally Automated Vehicles, Non-Driving-Related Tasks; Take-Over Requests; Text-Presentation Techniques; Audio-Assisted Reading

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1 INTRODUCTION

The Society of Automotive Engineers (SAE) outlines six levels of driving automation ranging from no driving automation (level 0) to full driving automation (level 5), i.e., requiring no human intervention [8]. Vehicles with level 3 automation or conditionally automated vehicles, do not require drivers' constant supervision, and allow them to engage in non-driving-related tasks (NDRTs) [42] (e.g., reading, watching a movie, or office work). However, the driver must remain receptive to requests to intervene, referred to as take-over requests (TORs) [11]. TORs are issued by the vehicle when the driving scenario is outside of the vehicle's operational design domain (ODD) or when the automated driving system (ADS) malfunctions [50].

While engaging in NDRTs could amplify safety risks to passengers and pedestrians, prohibiting NDRTs during automated drives could exacerbate safety risks by causing driver drowsiness [36], fatigue [61], and even worsen the take-over performance during long commutes [35]. Thus, a minimum level of NDRT engagement in conditionally automated vehicles would be required to maintain arousal and ensure adequate take-over performance [40]. Therefore, the design of user interfaces for supporting NDRTs involves critical trade-offs between safety and productivity: the driver must remain fallback-ready [43] while efficiently performing NDRTs, especially during prolonged NDRT engagement [36].

Reading is often the base of technological communication (e.g., email communication, instant messaging) which is commonly disallowed in SAE level 0-2 vehicles, but open to examination in SAE level 3 vehicles and above [39]. Studies have consistently identified reading as one of the more appealing NDRTs for automated vehicle drivers [22, 42, 54]. However, reading requires varying levels

of attention depending on the activity: for example, a driver may want to read instant text messages, which require short bursts of attention, or lengthier news articles requiring increased cognitive demand. Thus, we recognize the need for a thorough investigation into designing user interfaces that support reading as a non-driving task in conditionally automated vehicles.

One of the primary objectives of this study is to investigate the potential advantages of two prominent text-presentation techniques for enhancing reading capabilities and safety in conditionally automated vehicles (see Fig. 1). The first technique, vertical scrolling text presentation (VSTP) [3], involves the continuous vertical scrolling of multiple lines of text. By displaying several lines simultaneously, VSTP may reduce the working memory load [10]. Studies have notably shown VSTP can enhance reading speeds [27], potentially offering a more efficient reading experience within the visual environment. The second technique, rapid serial visual presentation (RSVP) [14], sequentially displays one word at a time at a fixed location. This approach not only reduces spatial footprint, thereby causing less visual obstruction, but also significantly minimizes eye movements during reading [41]. The combination of reduced visual obstruction and minimized eye movements in RSVP may enable the driver to maintain a stable peripheral vision and better identify objects and potential hazards in the driving environment, thereby enhancing their response to TORs. Furthermore, we investigate the integration of auditory speech displays with visual reading interfaces for NDRTs. This approach, in line with the multiple-resource theory [63], suggests that combining visual and auditory displays can enable more efficient information processing. This improved efficiency has the potential to reduce the cognitive workload of the user, an influential factor in take-over performance [38]. Additionally, this method affords drivers the flexibility to intermittently glance at the road, maintaining their productivity without compromising safety, potentially increasing both user preference and take-over performance.

Our goal is to evaluate the effectiveness of these two distinct text-presentation approaches in the context of conditionally automated vehicles. By comparing them, we aim to gain insights into their respective strengths and weaknesses for supporting reading as an NDRT. Thus, the following research questions guide our study:

- RQ1** What are the effects of VSTP and RSVP for reading on take-over performance, reading comprehension, cognitive workload, and user experience in conditionally automated vehicles?
- RQ2** How does the integration of auditory speech displays with VSTP and RSVP for reading affect take-over performance, reading comprehension, cognitive workload, and user experience in conditionally automated vehicles?

Our main contributions are as follows: **(C1)** An exploration of the effectiveness of two prominent text-presentation techniques — vertical scrolling text presentation (VSTP) and rapid serial visual presentation (RSVP) — for reading tasks in conditionally automated vehicles; and **(C2)** an investigation of the benefits and the implications of integrating auditory speech displays (ASDs) for reading in conditionally automated vehicles.

2 RELATED WORK

2.1 Display Modalities for Non-Driving-Related Tasks (NDRTs)

Conditionally automated vehicles present the intricate challenge of ensuring optimal cognitive resource allocation [21, 51]. Within this automation level, drivers have the opportunity to engage in non-driving-related tasks (NDRTs) but are expected to remain responsive to take-over requests (TORs). Lee et al. [28] highlighted that high cognitive loads during NDRTs might compromise a driver's take-over performance, leading to enhanced safety concerns. Oliver et al.'s [24] empirical studies corroborate these findings, indicating that NDRTs involving regular manual interactions with devices could detrimentally influence take-over capabilities.

To optimize drivers' ability to handle NDRTs during automated driving phases, researchers have delved into various display modalities. Head-up displays (HUDs) have been shown to reduce the time drivers spend alternating between NDRTs and roadway monitoring [56]. This approach results in faster NDRT completion [56, 57] and enhanced driving proficiency when compared to head-down displays [56, 59]. However, depending on the HUD's position, it may obstruct the driver's perception of road elements [13].

Auditory speech displays (ASDs) have been notably researched for SAE levels 0-2 vehicles [30, 32, 58], and their application in level 3 automated vehicles remains an area of ongoing research. These interfaces rely solely on audio, which could facilitate the driver to visually perceive the driving environment while engaged in an NDRT. ASDs could enhance safety by allowing the driver to be more receptive to ADS malfunctions that do not necessarily issue requests to intervene [8], a consideration that goes beyond the requirements of SAE level 2 vehicles. Schartmüller et al. [52] reported higher NDRT performance with HUDs compared to ASDs but found ASDs appealing to the users due to their potential to enable users to observe their surroundings during NDRTs.

We see a significant research gap regarding the potential benefits of integrating HUD and ASD for safety and productivity, based on multiple-resource theory [63]. This theory notably proposes the fusion of auditory and visual cues to facilitate efficient information processing [2]. In this study, we thus explore the integration of heads-up and auditory speech displays to support safe and productive reading in SAE level 3 vehicles.

2.2 Text-Presentation Techniques

Extensive research has been conducted to evaluate various text-presentation techniques in terms of text comprehension, mental workload, and eye movements. Saccadic eye movements, which involve positioning the eyes at the beginning of the next word to be read, have been identified as a limiting factor in reading efficiency [23, 33]. Reading comprehension requires complex lexical, syntactic, semantic, and contextual processing, often across multiple sentences [47]. Displaying multiple sentences at once could facilitate comprehension by allowing users to revisit previous sentences as needed [10, 34]. However, some studies have reported that text comprehension performance may not be significantly impacted when the amount of text presented is reduced to a single line [12].

A variety of text-presentation techniques have been developed to improve reading across diverse contexts. The most prevalent include page presentation, vertically scrolling text presentation (VSTP) [3], leading presentation (i.e., horizontally scrolling text) [19], and rapid serial visual presentation (RSVP) [14]. VSTP has been reported to lead to the fastest reading speed among various mobile displays when compared to RSVP and leading presentation, although its comprehension levels were found to be lower [27]. On the other hand, RSVP has been associated with fewer eye movements on mobile devices than VSTP and leading presentation [41]. Moreover, Juola et al. [26] observed that RSVP yielded higher reading accuracy than leading presentation on an eight-character-wide display.

Historically, text-presentation techniques have been predominantly studied with reading as the sole task. However, recent advancements in mobile technology have sparked interest in supporting reading within dual-task environments. Notably, a study on designing reading interfaces for smart glasses revealed that optimal designs are context-dependent: leading presentation outperformed RSVP when walking, but the opposite occurred in seated conditions [49]. Another study compared RSVP and paragraph presentation in a virtual reality environment, where participants were tasked with assigning text labels to corresponding paintings in a virtual museum exhibition [48]. RSVP resulted in lower error rates and shorter task completion times compared to normal page presentation.

Based on previous research, RSVP and VSTP emerge as two prominent techniques that enhance productivity across numerous contexts. Nevertheless, it remains unclear whether their performance extends to non-driving-related tasks (NDRTs) in conditionally automated vehicles. In this work, we thus evaluate the effectiveness of VSTP and RSVP in this setting, where drivers must maintain both adequate reading comprehension and the ability to regain situational awareness.

3 METHODOLOGY

We employed a 2×2 factorial within-subject design with two variables: text-presentation technique (TPT) with levels *VSTP* and *RSVP*, and audio assistance (AA) with levels *Disabled* and *Enabled* (i.e., whether the ASD is enabled). Each participant interacted with 4 different reading interfaces in a conditionally automated vehicle driving simulation.

3.1 Hypotheses

Based on our research questions outlined in Section 1, we postulate the following hypotheses:

RQ1 *Text-Presentation Techniques*

Reading using RSVP will lead to **(H1.1)** significant improvements in take-over performance (i.e., reaction time, longitudinal, and lateral control); **(H1.2)** higher comprehension scores; **(H1.3)** lower perceived cognitive workload; and **(H1.4)** higher user experience scores compared to VSTP.

RQ2 *Audio-Assisted Reading*

Audio-assisted reading will lead to **(H2.1)** significant improvements in take-over performance (i.e., reaction time, longitudinal, and lateral control); **(H2.2)** higher comprehension scores; **(H2.3)** perceived cognitive workload; and

(H2.4) higher user experience scores compared to non-audio-assisted reading.

3.2 Implementation

3.2.1 Driving simulator. We used a modified version of the open-source DReyeVR [55] simulator. The simulator ran on a high-performance PC: Intel i9-12900KF CPU, 32GB RAM, and an NVIDIA GeForce GTX 3080 graphics card. The simulation was displayed on a triple-monitor setup, each being a 27-inch monitor with 1080p HD resolution, 1 ms response time and 75Hz refresh rate. This resulted in a 120° horizontal field of view rendering at approximately 60 FPS. Participants were seated in a fixed-to-ground racing simulator cockpit 1.3 meters away from the monitors in a dark environment (black-painted room with 10 lux lighting). A Logitech G920 steering wheel and pedals were used to control the vehicle during manual driving, and noise-canceling headphones were worn to make the simulation more immersive.



Figure 2: Driving simulator setup with triple-monitor display, racing cockpit, steering wheel and pedals.

3.2.2 Driving conditions. Participants engaged in the study by operating a simulated conditionally automated vehicle, specifically a left-seat-driven Tesla Model 3, configured to maintain a maximum speed of 120 km/h during automated drives. Visibility was augmented through the left side mirror and the rear mirror, while driver input was systematically disabled upon engagement of the automated driving system (ADS). The virtual driving environment was designed to replicate moderate rain and midday conditions on an infinitely-looping three-lane highway, devoid of traffic hindrances. To maintain uniformity in the experimental conditions and facilitate a controlled comparison of participant responses, TORs were consistently issued at a predetermined distance of approximately 1200 meters (roughly 60 seconds) after the engagement of the ADS, specifically at the conclusion of straight road sections preceding a minor bend.

3.2.3 Head-up display (HUD). The text-presentation techniques were implemented to be displayed on a virtual HUD. The parameters for our HUD and text-presentation techniques were established based on findings from a pilot study. The HUD consisted of a white semi-transparent (50% opacity) background of varied size based on the text-presentation technique used: 19.5×4 cm ($8.6^\circ \times 1.8^\circ$ in

visual angle) for RSVP, capable of displaying 15-character words, and 23×10 cm ($10.1^\circ \times 4.4^\circ$) for VSTP, displaying 5 lines and a maximum of 25 characters per line. Despite suggestions for lower placement [44], we positioned the virtual HUD near the top of the windshield, aligned with the driver's seat and tilted 30° towards the driver's view, to accommodate VSTP's larger spatial footprint without obstructing the driver's line of sight and to standardize HUD positions across both techniques. The text was displayed on the HUD using a 52-point Arial Black (0.8° vertically) to maximize contrast and readability from a far distance (see A in Fig. 1).

3.2.4 Text-presentation interfaces. For the audio-enabled interfaces in this study, Microsoft SAPI 5.3¹, a text-to-speech software, was employed to provide audio assistance, utilizing an American English male speaker's voice. Both the text-presentation techniques and the accompanying audio assistance were calibrated to a reading rate of 180 words per minute. This rate was derived from the lower end of reading rates for non-fiction books [4] and further confirmed through a pilot study, aiming to accommodate the heightened cognitive demands associated with reading in a conditionally automated driving environment. The VSTP implementation displayed five lines of text simultaneously, with the third line highlighting the words vocalized by the text-to-speech software when activated. The text was programmed to scroll upward automatically, revealing subsequent lines in a timely manner. Conversely, our RSVP implementation presented only a single word at a time on the HUD. To ensure precise synchronization with the audio assistance, a dynamic time budget was allocated to each word displayed on the HUD, reflecting the pronunciation duration in Microsoft SAPI.

Additionally, we made adjustments to our RSVP interfaces to address the inherent advantage of VSTP: after a TOR was attended, the RSVP interface restarts at the beginning of the current sentence rather than the last shown word. This provided more context to the driver, allowing them to refocus on the reading task.

3.3 Tasks

During a trial, participants were instructed to engage in the reading task upon activation of the automated driving system (ADS), specifically before a take-over request (TOR) was issued and after the resumption of ADS following a TOR. It should be noted that the reading task was intentionally paused during the TOR to allow participants to respond. Simultaneously, participants were required to remain receptive to TORs issued by the vehicle as it approached the boundaries of its operational design domain (ODD).

3.3.1 Non-driving-related task (NDRT). Participants completed eight reading tasks. For each task, a text was selected from a pool of nine short reading comprehension texts (grade 11 level). Each text was presented individually on the HUD, with $M = 228$ and $SD = 16.14$ words. The additional text in the pool ensured continuity in case of a collision requiring an extra trial. To assess reading comprehension, after each trial, participants were asked to answer a set of multiple-choice questions pertaining to the text they had read. Each of the nine texts had either 4 or 5 multiple-choice questions (e.g., "What is the attitude of modern children towards school?", with options A) They think there is too much homework, B) They

¹Microsoft Speech API (SAPI) 5.3 (last accessed on October, 2023)



Figure 3: Four TOR scenarios: A) abrupt stoppage by a leading vehicle, B) entering a construction zone, C) foggy weather conditions, and D) animals crossing the road.

believe education is not just for people with a lot of money, C) They feel people should be prepared to pay for good schools, and D) They want to go to schools that were once only for the rich). The questions were designed to focus on the information presented and the overall themes of the text, ensuring that they would be easy to answer for participants who had read and comprehended the texts.

3.3.2 Take-over requests (TOR). Take-over requests (TORs) are triggered when the driving situation falls outside the vehicle's operational design domain (ODD). The ODD encompasses specific traffic, weather and lighting conditions, geographical restrictions, and various road features, such as lane markings and barriers [8]. TORs are visually indicated in red on the windshield and are accompanied by an auditory alarm throughout the take-over period, thereby providing a multi-modal alert to the driver (Fig. 3). When the scenario falls outside these parameters, the take-over procedure is initiated: The driver must perceive the environment, comprehend the driving scenario, assume physical control of the vehicle, and maneuver to a safe environment for ADS resumption. Participants are instructed to mimic typical driving behavior to elicit natural responses to TORs within the study's controlled environment.

TORs are issued approximately 60 seconds after the start of a trial, at a distance that allows a 5-second *time headway* before the vehicle reaches an obstacle. The term *time headway* refers to the time required to reach the obstacle's current position [16]. Our study included four take-over scenarios, adapted from [60] and shown in Fig. 3, to represent a range of events for SAE level 3 vehicles: A) *abrupt stoppage by a leading vehicle*, B) *entering a construction zone*, C) *foggy weather conditions*, and D) *animals crossing the road*. These scenarios require the driver to slow down by braking, with high adjacent-lane traffic simulation to discourage lane changes. We opted for a consistent driving response to allow comparisons between different conditions and to control for behavioral variability. Additionally, participants are instructed to maintain their current lane, deviating only when collision avoidance requires an imminent lane change.

The ADS is re-engaged when the conditions re-align with the ODD. This re-engagement occurs when the lead vehicle moves at regular speed again (A), when obstacles are passed or removed from the road (B, D), or when weather conditions improve (C). To clearly communicate this transition back to automated control, an alert is displayed on the windshield, stating, "Autopilot enabled; resuming reading task," thereby notifying the driver that control has been resumed by the ADS.

3.4 Measures

3.4.1 Take-over performance. Cao et al. [5] proposed a framework that assesses take-over performance based on two dimensions: *take-over time* and *take-over quality*. The *time* aspect focuses on the duration it takes to regain control of the vehicle after a TOR is issued, while *quality* refers to the overall quality of the driver's take-over maneuver.

To evaluate the time aspect, we measure the take-over reaction time (RT), which is determined as the time difference between the issuance of the TOR and the driver's intervention, either by rotating the steering wheel by at least 2° or applying the brakes with at least 10% effectiveness (based on the recommendation of Gold et al. [18]).

To assess take-over quality, our methodology incorporates metrics that capture both lateral and longitudinal driving performance. First, we measure the standard deviation of lateral position (SDLP), a widely reported metric for evaluating lane-keeping performance [5]. This metric consists of the standard deviation of the vehicle's offset from the center of its current lane during manual drive. We also measure the steering wheel reversal rate (SWRR), which quantifies the number of steering wheel reversals per minute, exceeding a predetermined minimum angular value, known as the gap size [31]. For our study, we set the gap size at 2° . Additionally, we record the maximum (MSWR) and standard deviation of steering wheel rotation (SDSWR) during manual driving to assess the potential risk of vehicle destabilization [45]. Finally, we calculate the average brake actuation (ABA) [52], as the average of all non-zero braking inputs provided by the driver.

3.4.2 Workload and user experience. We assess the perceived cognitive workload for each interface through the NASA-TLX questionnaire [20] on a 0-100 point scale, and user experience with the user experience questionnaire (UEQ-S) [53]. We additionally gather information on the driver's preferences regarding the text-presentation techniques, and audio-assisted reading in a conditionally automated vehicle setting by asking either/or questions through an end-of-study questionnaire.

3.4.3 Reading comprehension. We assess reading comprehension through the participant's accuracy in correctly answering multiple-choice questions at the end of each trial, based on the text presented during the trial.

3.5 Procedure

The procedure is summarized in Figure 4. Participants first complete a pre-study questionnaire gathering demographic information

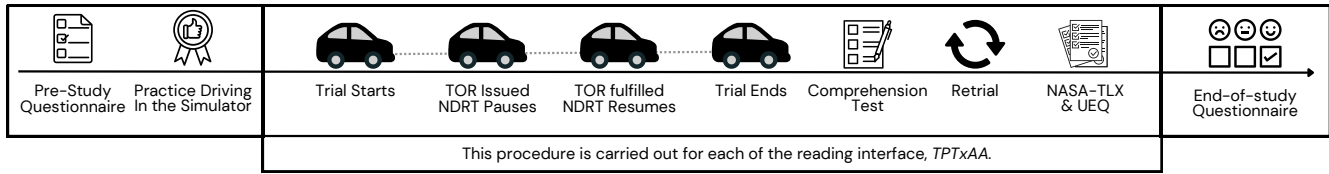


Figure 4: Overview of the study procedure.

as well as their driving experience. Participants then take as much time as needed to familiarize themselves with the simulator, learning about the HUD, TOR functionalities, and vehicle controls, and adjusting the volume to their comfort.

Every trial begins with the vehicle engaged with ADS and the reading task running on a specific interface displayed on the HUD. A TOR is issued after a certain time, prompting the participant to assume manual control of the vehicle. If a major collision occurs subsequent to a take-over request, i.e., the vehicle can no longer be driven on the road, the trial would restart with a new text. The trial then ends upon completion of the reading task on the HUD, and filling out the reading comprehension test. Each session with a participant consists of four blocks: one per each reading interface (2×2). Each block involves completing two consecutive trials with a particular interface, followed by filling the user experience questionnaire (UEQ-S) [53] and the NASA-TLX questionnaire [20] for the reading task through that interface. The session ends with a post-study questionnaire on the participant's experience during the study, as well as their preferences regarding the text-presentation techniques and audio assistance. Participants complete in total eight trials per session.

The interfaces were counterbalanced using a 4×4 balanced Latin square, while the four take-over scenarios were counterbalanced using an 8×8 balanced Latin square. Each scenario appeared twice to ensure equal distribution across participants. The presented texts were randomized, with each text appearing only once per participant.

3.6 Participants

32 participants (5 female, 27 male) aged between 17 and 55 ($M = 22.62$, $SD = 7.35$) were recruited. Participants were required to have a minimum of six months driving experience, demonstrate fluency in English, and have no serious motion sickness issues. The study took place at a post-secondary institution in North America and most of the participants were students at the institution. Participants reported having an average of ($M = 4.1$, $SD = 6.42$) years of driving experience, with 53.13% driving almost every day, 21.88% driving a few times a week, 15.63% driving a few days a month, and others driving infrequently. Their reported total kilometers driven ranged from 0-25,000 km ($f = 12$), 25,000-50,000 km ($f = 9$), 50,000-100,000 km ($f = 9$), and 100,000-200,000 km ($f = 2$). 37.4% of the participants were familiar with the RSVP technique. Participants received a \$25 Amazon e-gift card for their participation.

4 RESULTS

Each participant had two trials per reading interface. The means of the trials are used as take-over performance and reading comprehension scores. We performed two-way repeated ANOVA with

text-presentation technique and audio assistance as factors on all measures with Bonferroni correction, followed by paired t-tests for post-hoc analysis where relevant. We use Cohen's guidelines [7] to examine the effect sizes η^2 and Cohen's d , i.e., small effect sizes are $\eta^2 \geq 0.01$ and $d \geq 0.2$, medium are $\eta^2 \geq 0.06$ and $d \geq 0.5$, and large are $\eta^2 \geq 0.14$ and $d \geq 0.8$.

4.1 Take-Over Performance

In our study, we observed one major collision, which transpired during the use of VSTP without audio assistance. All the take-over scenarios implemented were designed to be critical braking scenarios, underscoring the necessity for an timely breaking response. To delve deeper into the drivers' ability to regain situational awareness following a TOR, we extended our analysis to include not only the overall driving performance, from the moment the TOR was issued until it was successfully attended to, but also the first five seconds of driving performance post-TOR. The take-over performance measures, corresponding to each of the 2×2 reading interfaces, are comprehensively summarized and presented in Table 1.

Regarding the time aspect of the take-over performance, an interaction effect emerged on reaction time, $F(1, 31) = 6.54$, $p = .02$, $\eta_p^2 = .17$ (see Fig. 5). Pairwise comparisons revealed an effect of audio assistance for the RSVP technique, $t(31) = 2.28$, $p = .03$, $d = 0.48$. Participants reacted quicker with audio assistance ($M = 1.06$, $SD = .27$) compared to no audio assistance ($M = 1.22$, $SD = .44$). No other effects were found.

For the quality aspect of the first five seconds of take-over performance, we observed a main effect of the text-presentation technique on ABA, $F(1, 31) = 4.49$, $p = .04$, $\eta_p^2 = .13$ (see Fig. 5). RSVP ($M = .19$, $SD = .10$) yielded significantly lower brake actuation than VSTP ($M = .22$, $SD = .15$). We also observed an interaction effect on ABA, $F(1, 31) = 4.30$, $p = .05$, $\eta_p^2 = .12$. The effect of the text-presentation technique emerged only when audio assistance was disabled: RSVP ($M = .18$, $SD = .08$) induced significantly lower brake actuation compared to VSTP ($M = .25$, $SD = .16$) when audio assistance was disabled, $t(31) = 2.86$, $p = .01$, $\eta^2 = .51$. Upon analyzing the overall take-over quality, we only observed statistically significant main effects for the SDSWR and MSWR measures (Fig. 5). Enabling audio assistance, $F(1, 31) = 5.44$, $p = .03$, $\eta_p^2 = .15$, resulted in significantly higher SDSWR ($M = 11.30$, $SD = 7.56$) than disabling it ($M = 9.16$, $SD = 4.03$). Lastly, enabling audio assistance, $F(1, 31) = 7.16$, $p = .01$, $\eta_p^2 = .19$, revealed significantly higher MSWR ($M = 40.05$, $SD = 23.00$) compared to disabling it ($M = 32.03$, $SD = 14.28$). No other effects were found.

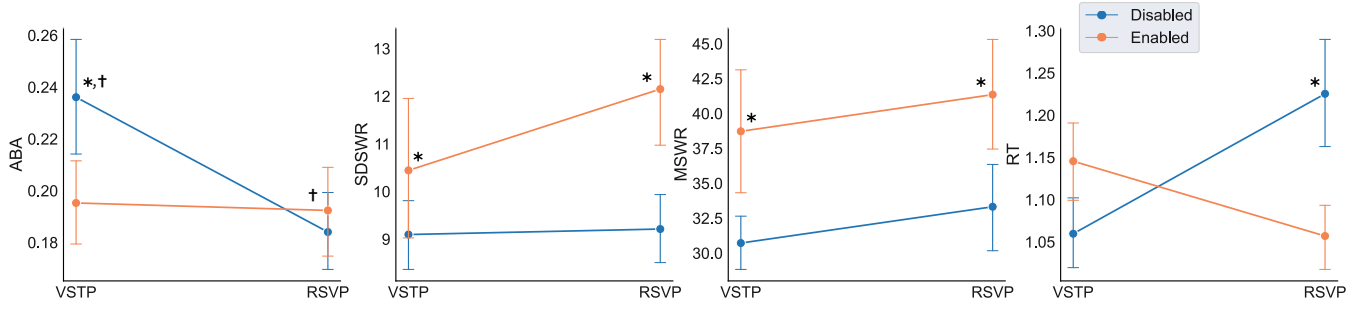


Figure 5: Line plot of take-over performance measures (from left to right): average brake actuation (ABA) during the first 5 seconds, overall standard deviation and maximum steering wheel rotation (SDSWR and MSWR), and reaction time (RT). * shows significance ($p < 0.05$) between disabled and enabled, and † shows significance ($p < 0.05$) between RSVP and VSTP.

Measures	VSTP-D	VSTP-E	RSVP-D	RSVP-E	VSTP vs RSVP		Disabled vs Enabled		Interaction	
					<i>F</i>	<i>p</i> (η_p^2)	<i>F</i>	<i>p</i> (η_p^2)	<i>F</i>	<i>p</i> (η_p^2)
RT (sec.)	1.06 (0.27)	1.15 (0.29)	1.23 (0.43)	1.06 (0.23)	.509	.481 (.02)	.804	.377 (.03)	6.538	.016 (.17)*
First 5 Seconds Take-over Performance										
SDLP (m.)	0.33 (0.13)	0.31 (0.1)	0.32 (0.1)	0.36 (0.13)	1.041	.316 (.03)	.401	.531 (.01)	2.183	.150 (.07)
SDSWR (deg.)	7.47 (3.41)	7.49 (3.49)	7.37 (3.95)	8.0 (3.51)	.086	.771 (.00)	.458	.504 (.02)	.246	.623 (.01)
MSWR (deg.)	21.68 (9.84)	21.78 (9.74)	20.94 (9.72)	23.32 (10.87)	.046	.832 (.00)	.732	.399 (.02)	.440	.512 (.01)
SWRR (rev./min.)	1.37 (3.73)	1.98 (5.18)	1.27 (5.98)	0.24 (0.66)	1.108	.301 (.04)	.081	.778 (.00)	1.235	.275 (.04)
ABA (%)	0.25 (0.16)	0.2 (0.13)	0.18 (0.08)	0.2 (0.11)	4.489	.042 (.13)*	.753	.392 (.02)	4.299	.047 (.12)*
Overall Take-over Performance										
SDLP (m.)	0.31 (0.13)	0.32 (0.08)	0.3 (0.09)	0.36 (0.12)	1.154	.291 (.04)	3.942	.056 (.11)	2.598	.117 (.08)
SDSWR (deg.)	9.1 (4.11)	10.44 (8.79)	9.21 (4.01)	12.15 (6.11)	.628	.434 (.02)	5.442	.026 (.15)*	.497	.486 (.02)
MSWR (deg.)	30.73 (11.15)	38.73 (25.08)	33.33 (16.91)	41.36 (21.03)	.510	.481 (.02)	7.155	.012 (.19)*	.000	.997 (.00)
SWRR (rev./min.)	7.45 (5.91)	9.29 (8.99)	6.85 (6.19)	6.59 (5.91)	1.685	.204 (.05)	.510	.481 (.02)	.735	.398 (.02)
ABA (%)	0.24 (0.15)	0.2 (0.11)	0.18 (0.08)	0.19 (0.11)	3.667	.065 (.11)	1.014	.322 (.03)	3.050	.091 (.09)

Table 1: (Left) Descriptive statistics and two-way repeated measures ANOVA statistics; (Right) Take-over performance measures ($df = 1, 31$). The appended -D and -E indicate that audio assistance is disabled and enabled, respectively. * indicates $p < 0.05$.

4.2 NASA-TLX and UEQ-S Ratings

All the scores except effort were significantly better when audio assistance was enabled compared to disabled (see Fig. 6 for descriptive statistics): perceived mental workload, $F(1, 31) = 12.50, p < .01, \eta_p^2 = .29$, physical workload, $F(1, 31) = 22.74, p < .01, \eta_p^2 = .42$, temporal demand, $F(1, 31) = 8.64, p = .01, \eta_p^2 = .22$, performance, $F(1, 31) = 7.28, p = .01, \eta_p^2 = .19$, and frustration, $F(1, 31) = 9.42, p < .01, \eta_p^2 = .23$. We also found a significant interaction effect on the perceived effort ratings, $F(1, 31) = 4.14, p = .05, \eta_p^2 = .12$. Pairwise comparisons revealed that RSVP with audio assistance had lower effort scores ($M = 43.13, SD = 24.03$) than VSTP with audio assistance ($M = 54.78, SD = 22.26; t(31) = 2.29, p = .03, d = .50$). No other effects were found.

In our analysis, we evaluated the user experience questionnaire (UEQ-S) ratings for both pragmatic quality (addressing ease of use and reliability) and hedonic quality (covering appeal and user satisfaction) for each interface, as depicted in Figure 7. A technical error resulted in the optional response to individual UEQ-S subscales, rather than mandatory as intended. We employed list-wise

deletion to balance each question's responses, resulting in sample sizes of $n = 31$ for pragmatic quality, $n = 28$ for hedonic quality, and $n = 32$ for overall quality. We found a main effect of audio assistance on pragmatic quality, $F(1, 30) = 34.66, p < .01, \eta_p^2 = .54$, hedonic quality, $F(1, 27) = 11.44, p < .01, \eta_p^2 = .30$, and overall quality, $F(1, 31) = 35.98, p < .01, \eta_p^2 = .54$. The scores all significantly favored enabling audio assistance. For text-presentation technique, we only found a significant main effect on the hedonic quality $F(1, 27) = 4.78, p < .04, \eta_p^2 = .15$. The hedonic quality was significantly better for RSVP ($M = 0.88, SD = 1.07$) than VSTP ($M = 0.41, SD = 1.30$). While we observed a general trend for pragmatic and overall qualities favoring RSVP, no significant differences were found. Furthermore, no interaction effects were observed.

4.3 Reading Comprehension

We found no significant effects of text-presentation technique and audio assistance, ($ps \geq .40$). Without audio assistance, comprehension scores were ($M = 66.46, SD = 17.19$) for VSTP and ($M = 69.14, SD = 16.40$) for RSVP. With audio assistance, scores were ($M = 68.18, SD = 19.72$) for VSTP and ($M = 65.81, SD = 18.59$) for RSVP.

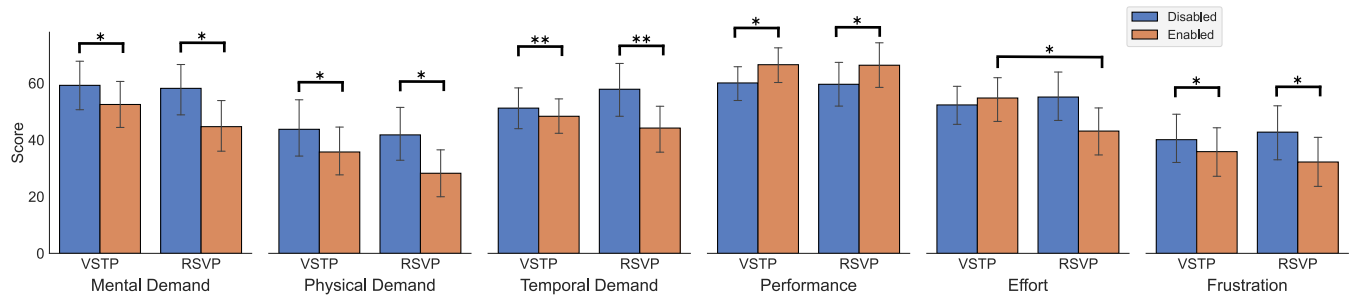


Figure 6: Bar plot of NASA-TLX scores for each reading interface. * indicates $p < 0.05$ and ** indicates $p < 0.001$.

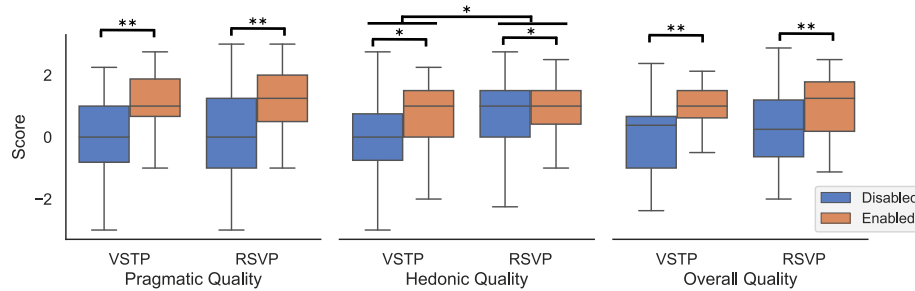


Figure 7: Box plot of UEQ-S quality scores for each reading interface. * indicates $p < 0.05$; ** indicates $p < 0.001$.

4.4 User Preference and Feedback

Responses from our questionnaires revealed a preference among participants for audio-assisted reading. Specifically, 27 participants (84.4%) favored enabling audio assistance, while only four (12.5%) preferred it disabled, and one (3.1%) expressed no preference. However, preferences concerning text-presentation techniques were more varied and showed no significant differences with audio assistance either enabled or disabled: without audio assistance, 17 participants (53.1%) opted for VSTP, 13 (40.6%) for RSVP, and two (6.3%) were indifferent; similarly, with audio assistance enabled, 18 participants (56.3%) chose VSTP, 13 (40.6%) RSVP, and one (3.1%) expressed no preference.

Many participants found that VSTP was more effective and simple to use: “VSTP was much better for both understanding and ease” (P04); “I prefer the scrolling text technique because it allows me to go back a little and understand anything [I] missed.” (P13). However, some participants also expressed a strong preference for the RSVP technique: “RSVP helped me focus on the text and better understand the content” (P05); “RSVP felt a whole lot better, maybe because it was easier to focus on one word at a time” (P08). Additionally, P11 emphasized that RSVP enabled them to understand the text better while comfortably observe the automated drive, and disliked VSTP as they felt it was “too much input for the brain”.

Participants suggested some enhancements to the text-presentation technique implementations used in the study ($f = 4$). For RSVP, P21 suggested displaying “few words instead of only one word [for RSVP] to better understand the context of the sentence.” One participant also suggested to highlight a specific letter to focus the mind on that

word and improve comprehension, as seen in the Spritz² reader. Regarding VSTP, one participant suggested “The [vertical] scrolling text could be more user friendly if it moved gradually, instead of moving line by line in rapid succession.”

Participants also commented on the various driving simulation functionalities which limited the realism of the simulation ($f = 4$). They observed that since the TORs were issued roughly at the same time in all trials, participants were occasionally able to anticipate when the TOR would be issued, and thus react faster (P03; P06; P22; P32). However, it was still difficult to anticipate “what the obstruction would be [as they were not in a predictable order]” (P22). Furthermore, the lack of a right-side mirror in the vehicle was noted to cause a “blind spot” (P13). Lastly, regarding the text-to-speech software used to provide audio assistance, one participant claimed it did not provide “appropriate pauses” (P12), and another recommended having both male and female voices (P30).

5 DISCUSSION

This study had two primary objectives: 1) to assess the effectiveness of two text-presentation techniques, rapid serial visual presentation (RSVP) and vertical scrolling text presentation (VSTP), in facilitating reading tasks in conditionally automated vehicles; and 2) to investigate the potential benefits of augmenting these techniques with audio assistance. The results offer insights into the applications of text-presentation techniques and audio-assisted reading within the specific context of automated driving, highlighting potential benefits and considerations. In the subsequent sections, we will discuss the main findings of our study, examine the limitations of our

²Spritz reader (last accessed on October, 2023)

methodology, and outline potential directions for future research in this area.

5.1 RQ1 — Text-Presentation Techniques

Both VSTP and RSVP emerged as initially promising text-presentation techniques for NDRT interfaces within the context of conditionally automated vehicles. VSTP, which displays multiple lines of text simultaneously, offers the advantage of enabling drivers to revisit previous sentences. This capability could potentially alleviate the working memory load, as suggested by previous research [10]. On the other hand, RSVP has been found to significantly reduce eye movements [41] and cause less visual obstruction, which could potentially improve the take-over performance.

We hypothesized the use of RSVP over VSTP would lead to better take-over performance, lower perceived workload, and higher reading-comprehension scores. However, we found no significant difference for any driving performance metric on overall take-over performance. Upon analyzing the initial take-over performance, i.e., the first 5 seconds, we observed a significant effect of the text-presentation technique on the average brake actuation (ABA): Compared to RSVP, VSTP led to significantly higher ABA with a moderately large effect size ($\eta_p^2 = .126$). Gold et al. [18] assert that high braking is a strong indicator of insufficient time for successful maneuver execution and a lack of situational awareness, which in turn affects take-over performance [6]. Furthermore, increased deceleration may heighten the risk of rear-end collisions and vehicle destabilization [46]. This outcome may be due to VSTP causing greater visual obstruction, hindering the driver's ability to identify hazardous objects beforehand, and simultaneously presenting an abundance of information, potentially overloading the user, as suggested by previous research [2]. Consequently, drivers may resort to higher braking as a means of buying time to understand the situation, resulting in deteriorated longitudinal control. Based on our findings, we conclude that using RSVP leads to better take-over performance, and thus, we **accept H1.1**.

Our analysis of the effort scores from NASA-TLX indicates that the text-presentation technique had a significant effect when audio assistance was enabled: Participants perceived using RSVP required less effort to complete the reading task compared to VSTP. This effect is likely due to VSTP displaying multiple lines simultaneously, which may hinder the fusion of audio and visual information required for reading, and impose additional cognitive processing [62]. In contrast, RSVP displays the current spoken word on the display in perfect synchronization. However, we did not find significant differences in the other NASA-TLX scores, which lead us to **reject H1.3**. Regarding the user experience, we found that RSVP yielded significantly higher scores than VSTP on the hedonic quality of the reading interface, i.e. users found RSVP more appealing and satisfying. Since we found no statistical difference in pragmatic and overall qualities, we can only **partially accept H1.4**. Finally, we also found no significant effect of text-presentation techniques on comprehension levels, with both techniques averaging 67% score. Thus, we **reject H1.2**.

In this study, we identified that the RSVP technique for reading in conditionally automated vehicles enhances the ability to regain situational awareness, thereby improving take-over performance,

and increasing hedonic quality. Our results indicate that RSVP may be a preferable method for reading interfaces in such vehicles, contributing to both enhanced take-over performance and user experience. While these findings provide valuable insights into the potential benefits of RSVP, further research is essential to validate these results in diverse driving conditions and reading contexts.

5.2 RQ2 — Audio-Assisted Reading

We hypothesized that incorporating audio assistance for reading could enable more efficient information processing, reduce cognitive workload, and enhance take-over performance. This approach was also expected to allow drivers to occasionally glance at the road, potentially improving both user preference and overall performance.

We found the perceived workload to differ significantly based on audio assistance: Perceived mental and physical workload, temporal demand, and frustration ratings were lower when audio assistance was enabled. Our findings thus suggest that incorporating audio assistance makes the reading task easier to process, which is important since human cognitive capacity is limited [37]. Furthermore, this supports the multiple-resource theory [63], which suggests that aligning audio and visual information enhances the efficiency of cognitive resource allocation and facilitates the processing of information [2]. In their meta-analysis, Wickens et al. [62] found that augmenting visual stimuli with auditory information in visual tasks enhanced communication-related task accuracy. While there was no significant effect of audio assistance on reading-comprehension scores, NASA-TLX performance scores suggest that participants perceived the audio assistance helpful for their NDRT performance. Thus, we **accept H2.3**. As for user experience, providing audio assistance for reading had a positive effect on both pragmatic and hedonic qualities, suggesting reading to be more engaging and effective when coupled with audio assistance during automated drives. The post-study questionnaire further revealed participants' preference for audio assistance for reading. We thus **accept H2.4**.

In terms of take-over performance, we found mixed results for the effect of audio assistance. Regarding the take-over time aspect, we found a medium-sized effect of audio assistance with RSVP ($d = .48$), where augmenting RSVP with audio assistance led to lower reaction times. We, however, focus our analysis on take-over quality, as it has been shown to serve as a relatively better indicator of take-over performance [24]. Furthermore, relying solely on drivers' reflexive responses is insufficient for measuring safety levels [15, 64]. We observed the maximum (MSWR), and standard deviation in steering wheel rotation (SDSWR), which describe the smoothness of a take-over maneuver [29], to be significantly higher when audio assistance was enabled for reading. This observed deterioration in lateral control might be attributed to increased engagement in NDRT with audio assistance. Previous research has shown that increased engagement can lead to the driver's gaze being directed away from the road less frequently, consequently resulting in diminished situational awareness [17]. Furthermore, our results align with previous studies on self-interruption, suggesting that mental and physical demands have a positive correlation to the prevalence of self-interruptions [1, 25]. Therefore, while productivity remains a key consideration in NDRT interface design, the

creation of interfaces that encourage task-interleaving, especially at ODD boundaries, during automated drives is essential for ensuring adequate take-over performance. Since we found no evidence suggesting that audio-assisted reading interfaces lead to improvements in take-over performance, we **reject H2.1**. In addition, our results suggest that audio-assisted reading had no significant impact on comprehension levels, and therefore we **reject H2.2**.

In our exploration of audio-assisted reading in conditionally automated vehicles, we discovered that this approach can reduce perceived cognitive workload and enhance the user experience. Although the impact on take-over performance was mixed, the positive effects on user engagement and efficiency are promising. These findings contribute to our understanding of how audio assistance can be integrated into reading interfaces within automated driving contexts.

5.3 Limitations and Future Work

Our study was conducted within a driving simulator, presenting certain limitations that might affect the authenticity of the driving experience, particularly due to the absence of haptic feedback and a real-world driving environment. Additionally, the design of the reading interfaces, which includes factors such as display transparency, synthetic voice tone, and reading speed, might not have been optimal for every participant, potentially introducing bias. With respect to our trial design, we acknowledge that having a constant interval before each TOR might have led to an unintended learning effect, despite our efforts to counterbalance the conditions. Moreover, the generalizability of our findings is constrained not only by the limited scope of take-over scenarios and the absence of eye-gaze data to assess participants' interruption behavior, but also by the relatively young mean age of our participants. Finally, the selected topics of the texts and the simulated conditions for the reading NDRT might not accurately represent real-world scenarios, such as a driver's lack of interest for certain articles or unforeseen weather changes.

In future research, we aim to evaluate reading comprehension using the reading-span task [9], as our current approach may overly depend on participants' memorization skills. We also plan to examine an auditory speech display (ASD)-only condition, in conjunction with the previously studied conditions, while collecting eye-gaze data. This approach will allow us to specifically understand the influence of integrating ASDs with visual displays on the driver's glance behavior, e.g., how often they interrupt the NDRT to gaze at the road. Finally, we intend to incorporate a broader array of critical and planned take-over scenarios, including lane changes and highway exits, to further generalize our findings to real-world scenarios.

6 CONCLUSION

This study has shed light on the potential of text-presentation techniques, specifically RSVP and VSTP, in the context of conditionally automated vehicles. Our findings highlight the effectiveness of RSVP in regaining situational awareness and user experience, while also revealing the promising role of audio-assisted reading in reducing cognitive workload. Although the implications of audio assistance on take-over performance require further exploration,

the overall positive impact on user engagement and efficiency is an encouraging step forward. Despite the limitations inherent in the simulated environment, our research offers valuable insights and paves the way for future innovations in interface design for reading NDRTs in automated driving. The potential to enhance both safety and productivity through these techniques underscores the importance of continued research and development in this emerging area.

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